

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element H Initial Template

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Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1 ex	Water must flow steadily and consistently from the IV bag source through the pipes to the plants.	1)) Set up the tubing and IV needle system on the PVC frame. 2) Attach a temporary water source (we used bottles, instead of the lv bag, just to pour the water. 3) Held the frame of the system b y hand(We did not come up with a way to support the system just yet) 4)Adjust the lv needle pressure to control the water flow. 5)) Watch if water reaches the ground through the holes in the PVC.	a. 3 trials due to setup challenges	Water flows through the tubing and out of the PVC holes with minimal leaks that are fixable. Water reaches the plants after adjusting the IV needle pressure. The water source was not stable during testing.	Fail: Water drips and leaks through the tubing, not too much, but not as much water reaches the plant, and also causes wasted water. Pass: IV needle pressure stays consistent and water reches plants. Goal: Fix all leaks and create a stable frame, and obtain a reliable water source to store the water in so that it flows through the tube without leakage
2 ex	Durable and Portable	1) Carry the system around the classroom carefully and set it down in	35 trials in different locations	System holds together with no	Fail: The structure could not support

		different locations. 2) Gently shake or bump the structure to test durability. 3) Inspect for any loose parts or breaks.		major breaks or falls apart. Some parts may shift but can be easily fixed.	itself and would fall over without being held. Pass: No major damage; can be moved without issues. Goal: Strong enough to be moved around regularly without breaking. As well as stand on its own with out falling.
3	Realistic to Build and Use	1) Assemble the system using only classroom-available materials. 2) Ask other students to try setting it up with instructions. 3) Time how long it takes to adjust the needle pressure.	3 trials with different users	Setup is completed within 8- 10 minutes. But once the needle pressure is found, it will not need to be adjusted again	Fail: Users struggle too much or system doesn't work after setup. Pass: Setup is doable with minimal guidance. Goal: Easy to build and maintain using simple tools and supplies.
4	2. Water Supply	1) Pour water manually into temporary water source (IV bag missing). 2) Adjust IV needle pressure to control water drip. 3) Observe water flow and check if it	3 trials (pressure changes each time)	Water flows through with minimal, fixable leaks. Water reaches plants with	Fail: Water leaks too much or does not reach the plant area. Pass: Water flows

		reaches the plants. 4) Watch for leaks and flow rate.		the needed needle pressure adjustments. The water source is not sturdy.	at a steady pace and reaches plants with stable pressure. Goal: Fix leaks and create a reliable, steady water source.
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Feedback From Field Experts (H5)

Mr. Butler has extensive experience in engineering design processes and He reviewed our testing plan and offered valuable suggestions to improve its accuracy and effectiveness.

- Recommended testing the IV tubing and PVC system under different water pressures to ensure it won't leak or clog under varying flow rates.
- Advised testing how long the system takes to deliver water from the source to the plant beds to measure efficiency.
- Emphasized the importance of documenting any unexpected outcomes during testing.

Revised Testing Details That Incorporate Field Expert Feedback (H5)

What is being tested?	Why is it being tested?	How will it be tested?	Criteria for Success
Water flow through IV tubing and PVC pipes	To ensure consistent and efficient water delivery to plants	Test the system under low, medium, and high water pressure using adjustable flow. Observe water output and check for leaks or clogs.	Water flows consistently at all pressure levels without leaks or blockages.
Soil moisture levels after irrigation	To ensure plants receive adequate water without overwatering	Use a soil moisture meter before and after irrigation. Repeat this test under different water pressures and record readings.	Moisture levels fall within the optimal range for plant health (neither too dry nor saturated).
Time it takes water to travel	To assess the efficiency and	Use a stopwatch to time water from IV bag entry	Water exits through holes within 1

from source to exit holes	responsiveness of the system	to pipe exit during multiple pressure trials.	minute of starting flow.
Durability of system under simulated rain conditions	To test the system's performance in outdoor-like conditions	Use a hose and timer to simulate rain in 3 cycles. Observe system stability and performance during and after simulation.	System remains stable and continues to function properly after repeated exposure.
Effectiveness of watering over time	To evaluate long-term performance and plant health	Test on different plant types. Monitor plant condition and soil moisture for 3–5 days. Record any visible issues or changes.	Plants show signs of healthy growth, no wilting or yellowing, and soil stays consistently moist.